

Student Performance Prediction System– Project 2

AI/ML Internship Project Report

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Student performance prediction is one of the practical applications of Machine Learning in the education sector. It helps estimate students' academic performance based on factors such as attendance, study hours, and internal marks.

This project uses a Linear Regression model to predict the final marks of students using historical academic data.

Objectives

- Learn data preprocessing techniques.
- Build a Machine Learning prediction model.
- Predict students' final marks. □ Visualize academic data.
- Evaluate model performance.

Tools & Technologies

Programming Language

- Python

Libraries

- Pandas
- Scikit-learn
- Matplotlib

Platform

- Google Colab

Algorithm

- Linear Regression

Dataset

The dataset contains the following attributes:

Attendance	Study Hours	Internal Marks	Final Result
95	8	48	95
90	7	45	90
85	6	42	85
80	6	40	80
75	5	38	75

70	4	35	70
65	4	30	65
60	3	28	60
55	2	22	50
50	1	18	40

The data is stored in a CSV file named **student_data.csv**.

The project follows these steps:

1. Load the student dataset.
2. Preprocess the data.
3. Split the dataset into training and testing sets.
4. Train the Linear Regression model.
5. Accept user input.
6. Predict final marks.
7. Visualize data using graphs.
8. Evaluate model accuracy.

Features

- Student Performance Prediction
- CSV File Handling
- Data Preprocessing
- Machine Learning Model
- Data Visualization
- Model Accuracy Evaluation
- Interactive User Input

The project uses **Linear Regression** algorithm.

Input Parameters

- Attendance
- Study Hours
- Internal Marks

Output

- Predicted Final Marks

The model identifies relationships between these academic factors and estimates the student's expected final marks.

Result

The system successfully predicts students' final marks based on user input.

Sample Input

Attendance = 88%

Study Hours = 6 Hours

Internal Marks = 43

Sample Output

Predicted Final Marks = 86.54

The project also generates a visualization graph showing the relationship between attendance and final performance and displays the model's accuracy score.

Future Scope

- Develop a graphical user interface (GUI).
- Use larger and real-world educational datasets.
- Implement advanced machine learning algorithms such as Random Forest or XGBoost.
- Add support for predicting grades and identifying at-risk students.

The Student Performance Prediction System demonstrates how Machine Learning can be applied to educational data. By using Linear Regression, the system predicts students' final marks based on attendance, study hours, and internal marks. The project also includes data visualization and model evaluation, making it a practical example of an AI/ML application.